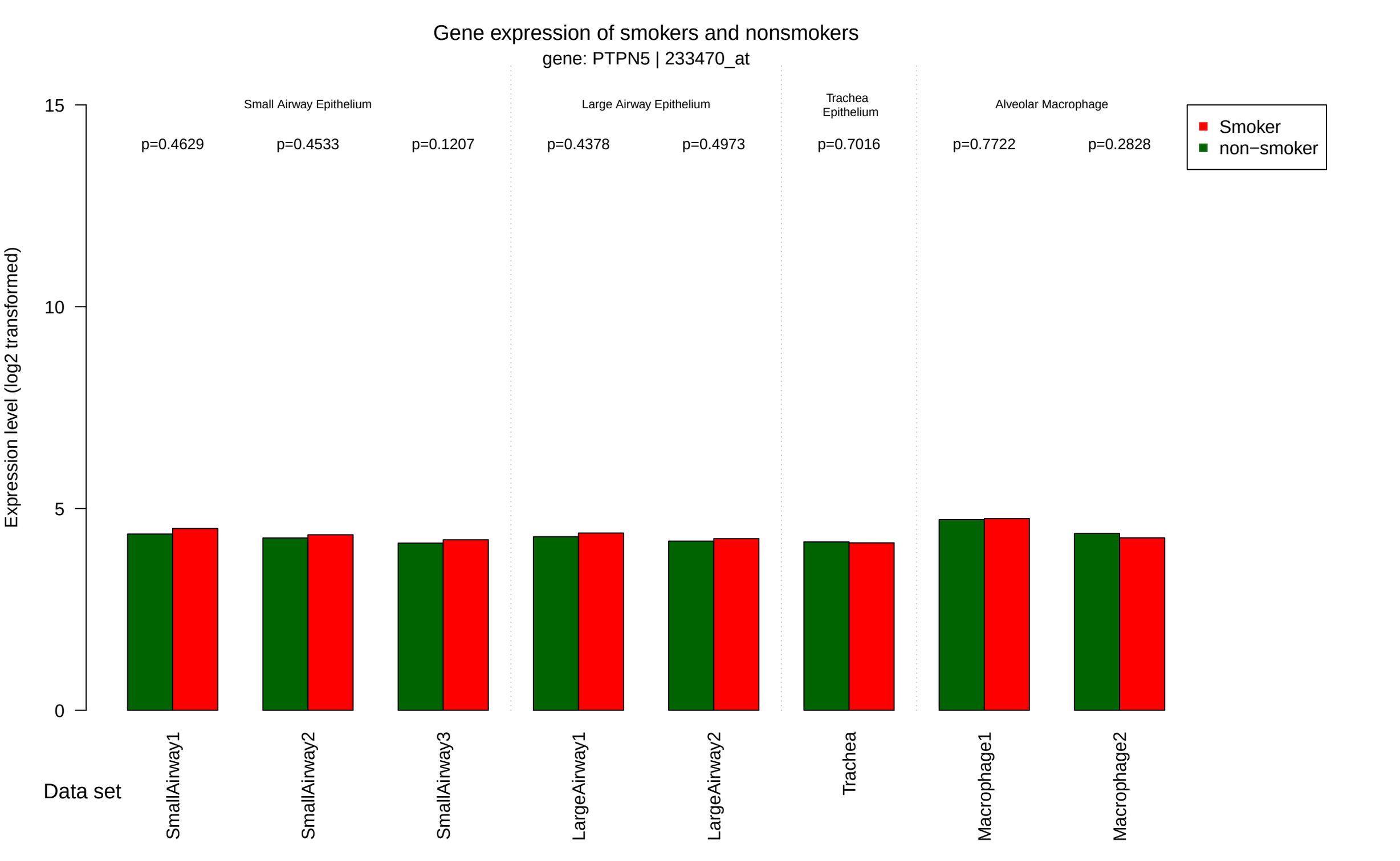
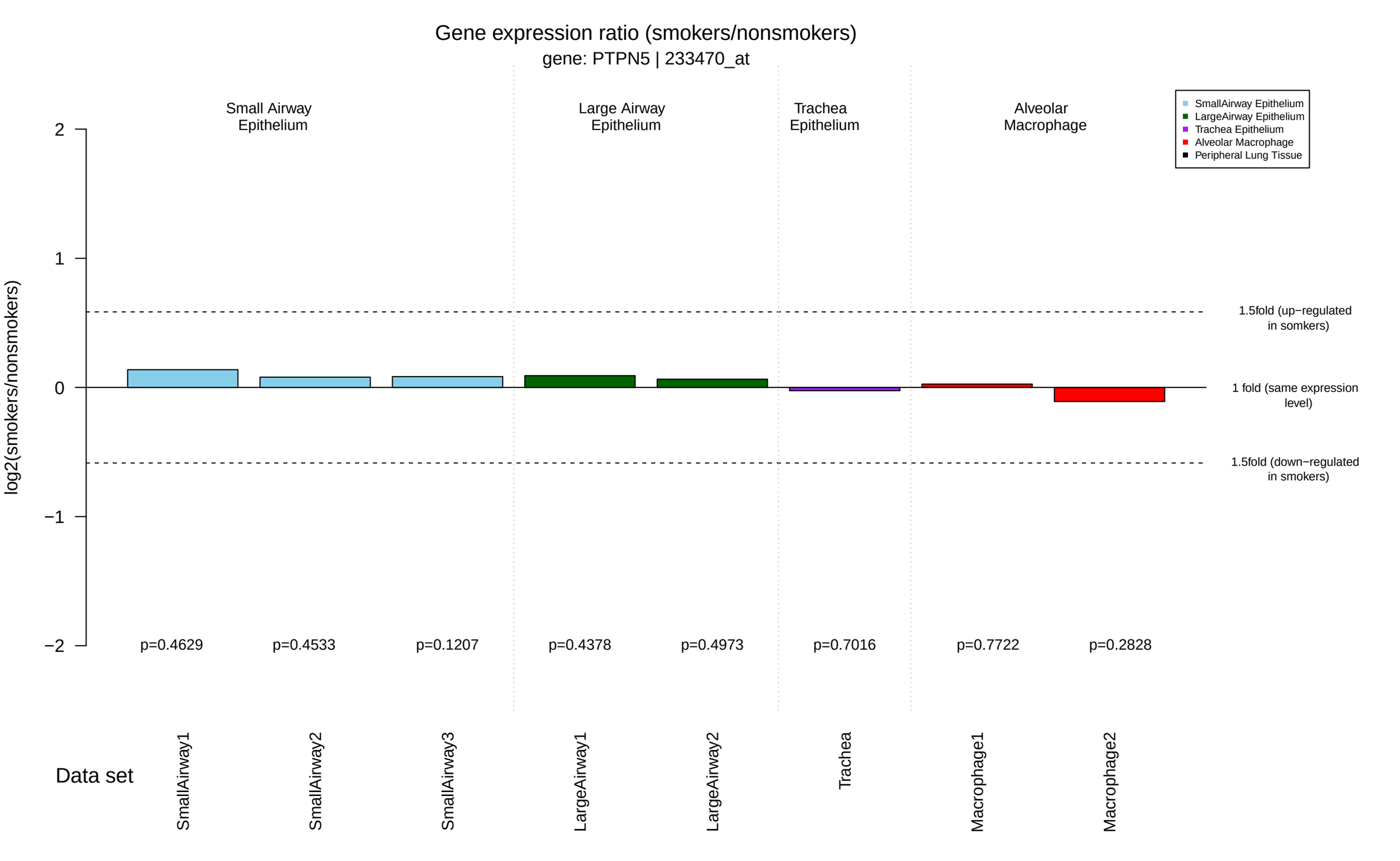
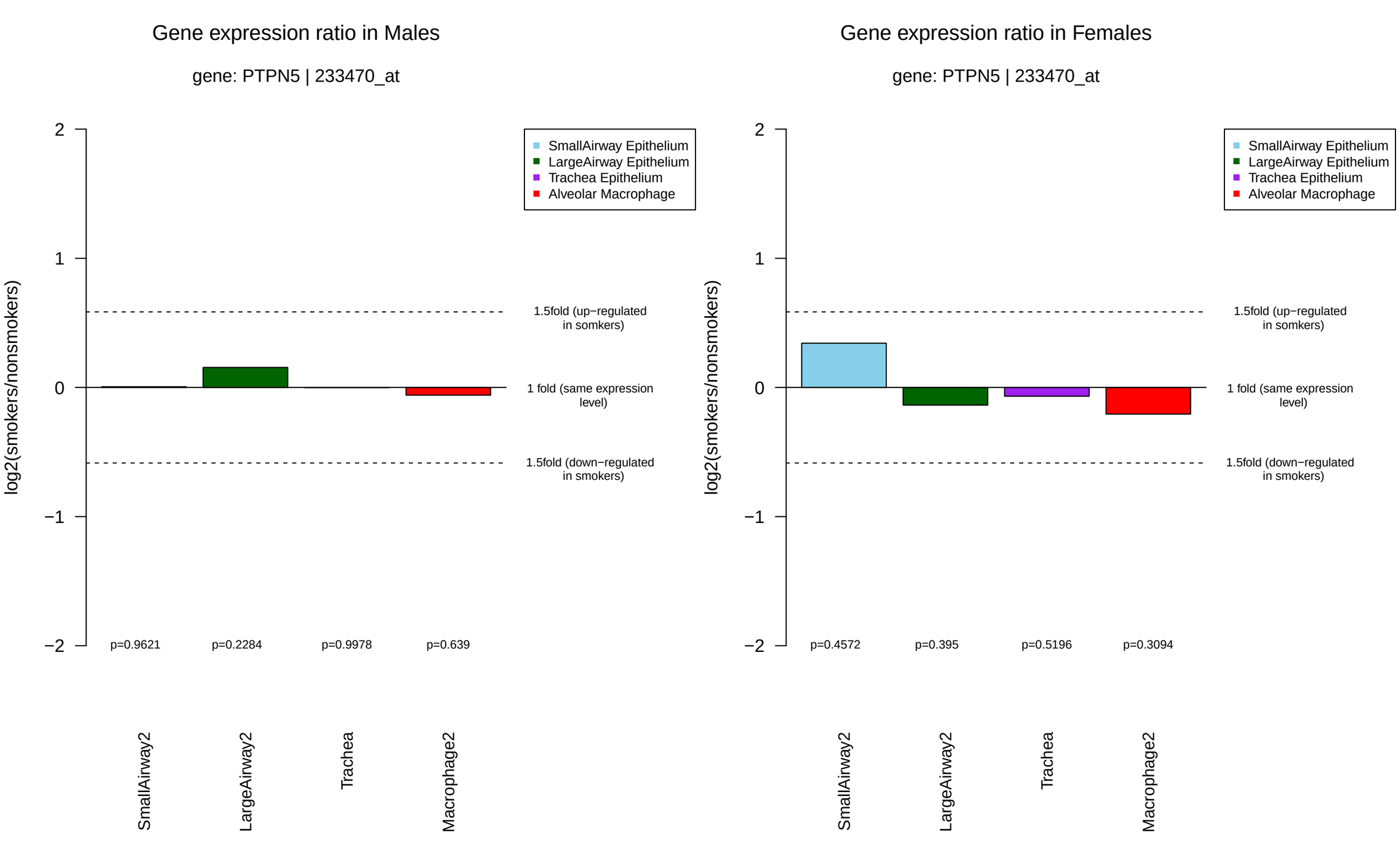
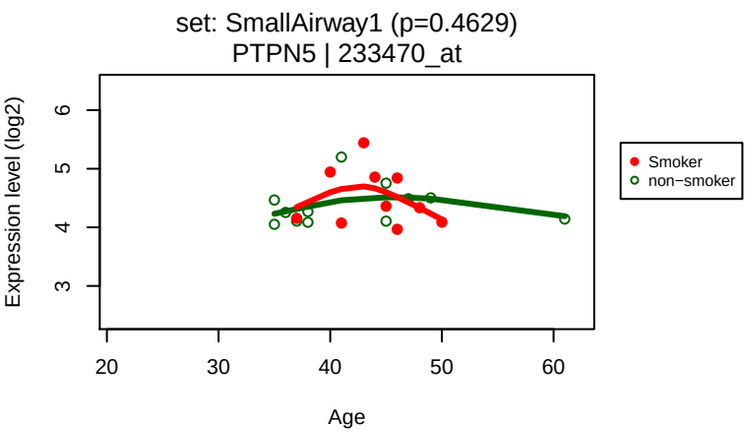
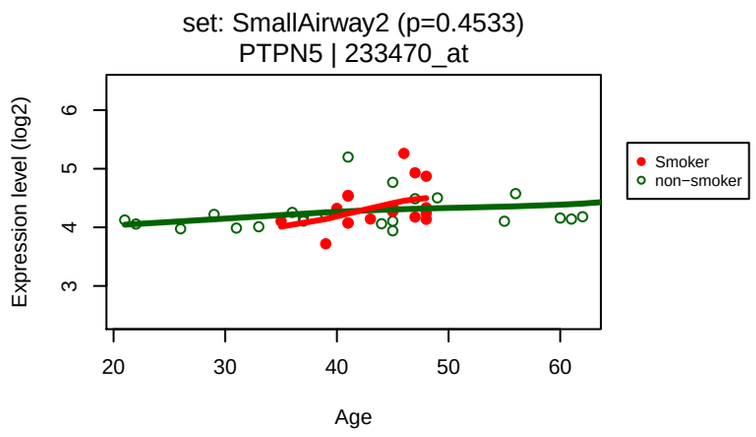
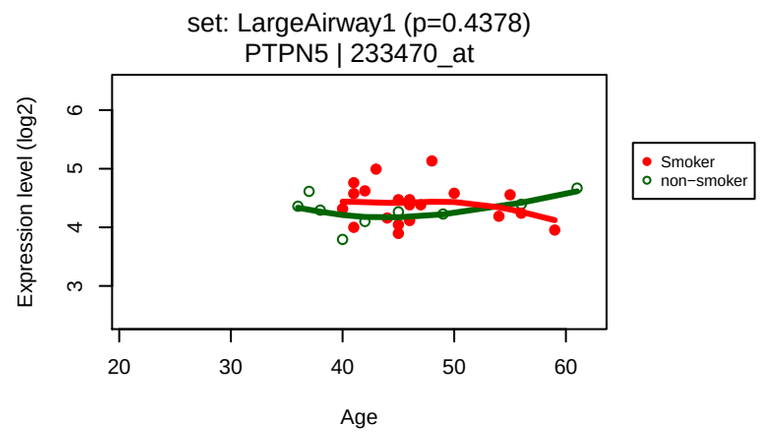

Gene expression of smokers and nonsmokers
gene: PTPN5 | 233470_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4629
p=0.4533
p=0.1207
p=0.4378
p=0.4973
p=0.7016
p=0.7722
p=0.2828
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

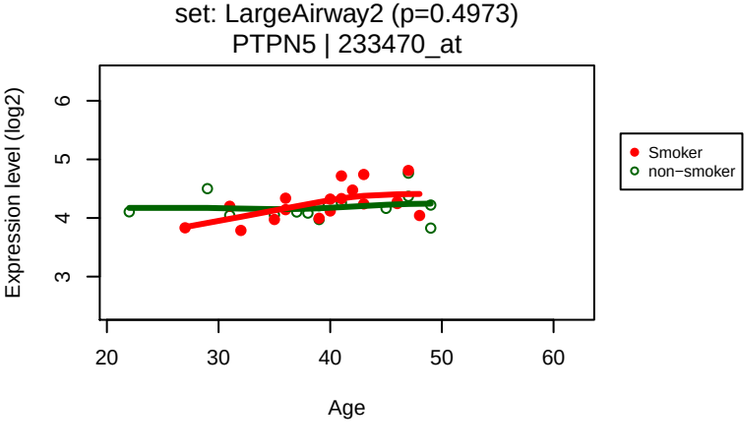

Gene expression ratio (smokers/nonsmokers)
gene: PTPN5 | 233470_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4629
p=0.4533
p=0.1207
p=0.4378
p=0.4973
p=0.7016
p=0.7722
p=0.2828
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

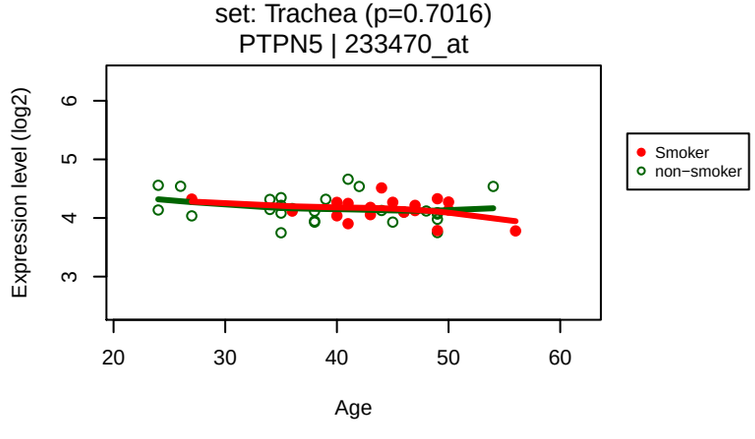

Gene expression ratio in Males
gene: PTPN5 | 233470_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9621
p=0.2284
p=0.9978
p=0.639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN5 | 233470_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4572
p=0.395
p=0.5196
p=0.3094
SmallAirway2
LargeAirway2
Trachea
Macrophage2

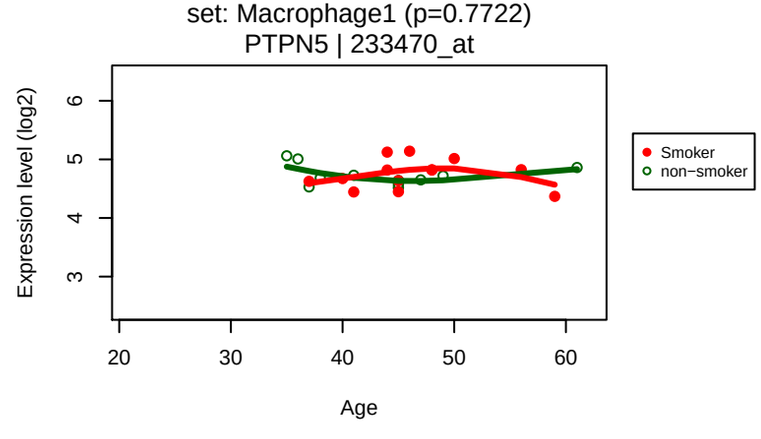

set: SmallAirway1 (p=0.4629)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker

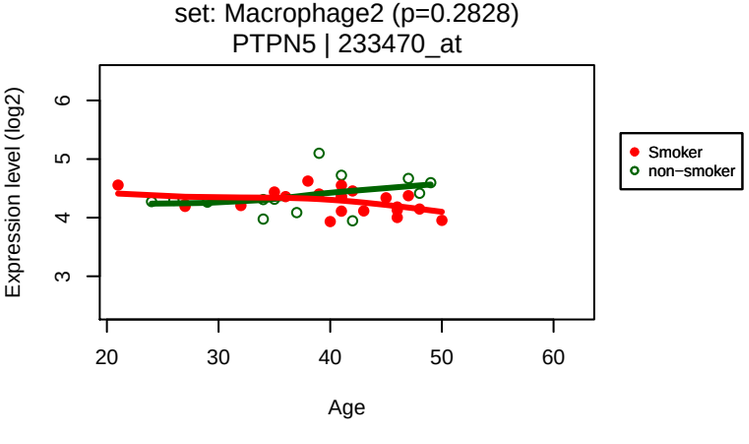

set: SmallAirway2 (p=0.4533)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker

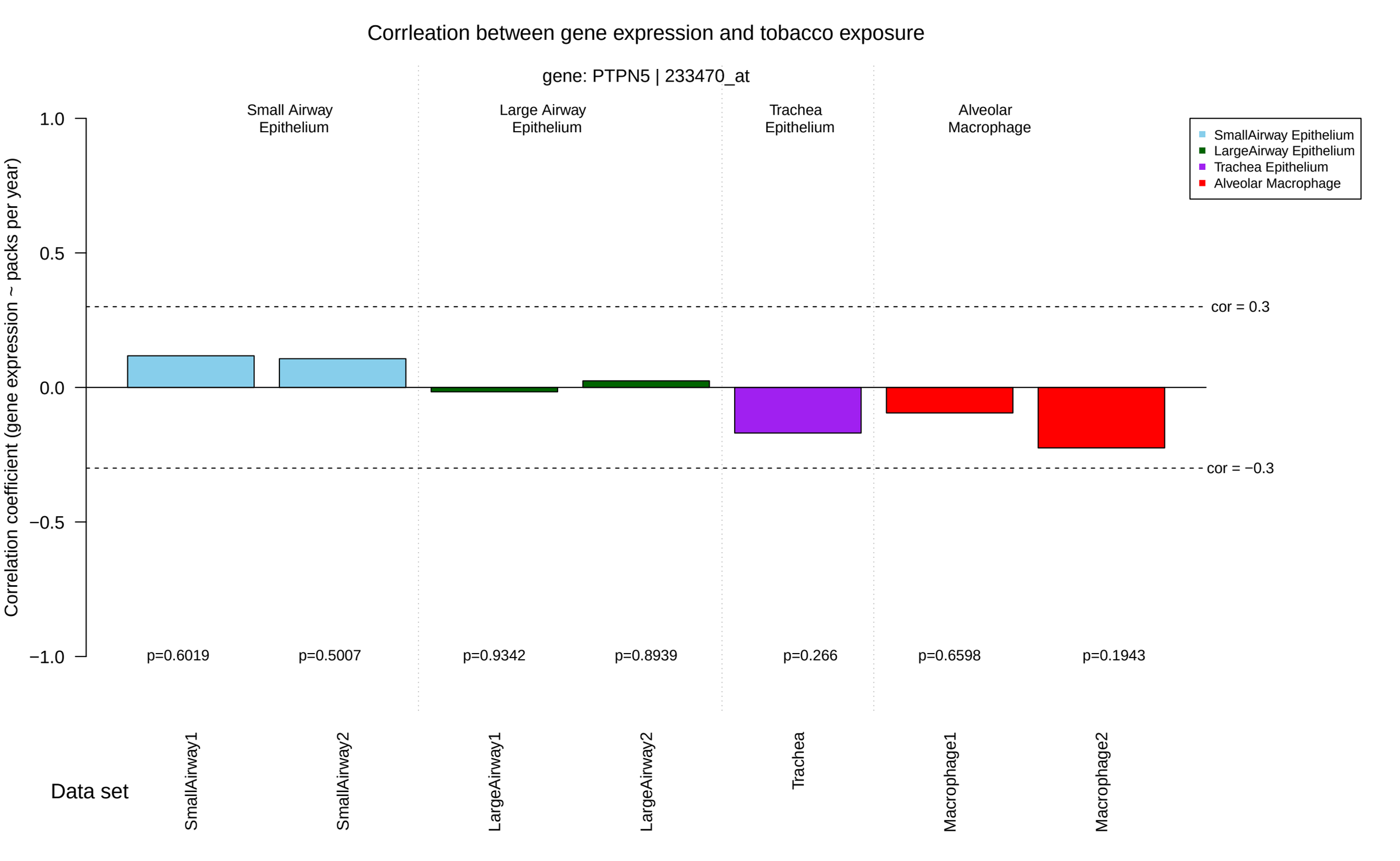

set: LargeAirway1 (p=0.4378)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4973)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7016)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7722)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2828)
PTPN5 | 233470_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN5 | 233470_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6019
p=0.5007
p=0.9342
p=0.8939
p=0.266
p=0.6598
p=0.1943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2